# KevlarGrid Explicit Solver Executive Report

Dynamic Mass-Spring Explicit Integration Simulation Summary — Powered by VibeDynaLITE

**OUTCOME: FAIL (PERFORATED)**

## 1. Impact Simulation & Structural Summary

### Fabric Configuration

**Material preset**: Kevlar 29
**Young's Modulus**: 71.0 GPa
**Failure Strain Limit**: 0.036 m/m
**Grid Nodes (Nx × Ny)**: 100 × 100
**Grid Spacing dx**: 0.0025 m
**Analysis Mode**: Mode A (Sizing Multiplier)
**Number of Plies**: 20 plies

### Striking Projectile

**Initial Mass**: 1.7 kg
**Initial Velocity Vz**: 100.0 m/s
**Live Kinetic Energy (KE)**: 8500.00 Joules
**Blade Profile Width**: 0.1524 m
**Edge Thickness**: 0.0127 m

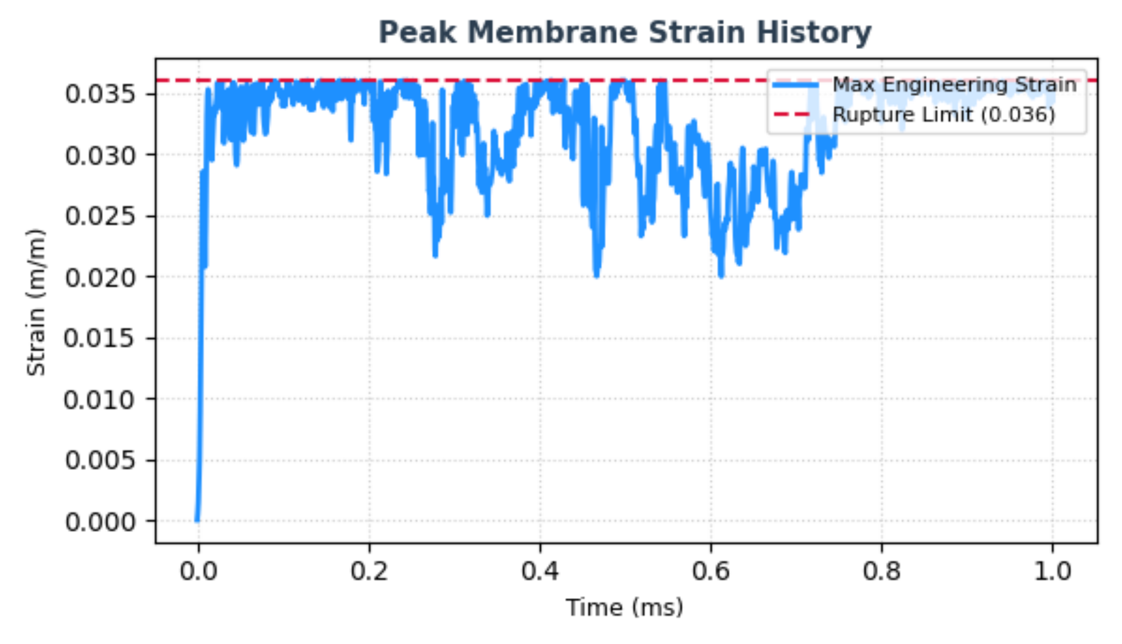
## 2. Solver Performance Telemetry

| Outcome Metric | Value | Standard Limits / Pass Conditions |
|---|---|---|
| **Projectile Arrest Outcome** | **FAIL (PERFORATED)** | MEMBRANE CONTAINMENT SUCCESSFUL |

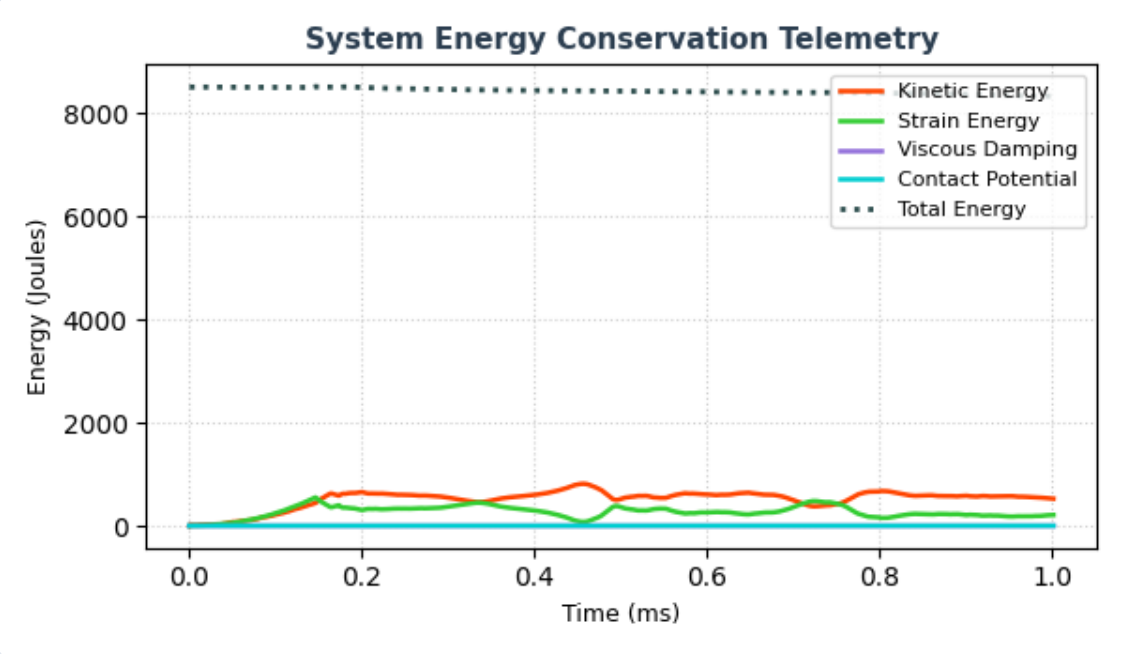
| Outcome Metric | Value | Standard Limits / Pass Conditions |
|---|---|---|
| **Peak Projectile Deceleration** | 34727.95 G's | Dynamic structural loading cap |
| **Yarn Springs Rupture Ratio** | 8.50% | Fabric structural degradation metric |
| **Residual Velocity** | 93.83 m/s | Exit velocity post-perforation |
| **Maximum Ply Perforation Index** | 0 / 20 | Total snap-through layer sequence |

## 3. Structural Telemetry Graphs

### Peak Fabric Strain Time History



## System Energy Balance Conservation

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